# 📘 Day 24 — ReAct Prompting & Structured Prompting

## 🚀 Introduction

In previous lectures, we learned:

- Prompt Engineering
- Zero-Shot / Few-Shot Learning
- Chain-of-Thought (CoT)

Now the next evolution is:

### 👉 ReAct Prompting

Modern AI systems do not just:

- generate text
- answer questions

They can now:

- reason
- use tools
- interact with systems
- retrieve information
- perform actions

This capability is powered by:

### 🧠 ReAct Prompting

Along with this, modern production AI systems also require:

### 👉 Structured Prompting

because enterprises need:

- reliable outputs
- predictable formatting
- automation-friendly responses

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## 🔷 1. What is ReAct Prompting?

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### ✅ Definition

ReAct stands for:

### 👉 Reason + Act

It is a prompting technique where the model:

1. reasons about the problem
2. decides an action
3. performs the action using a tool
4. observes the result
5. reasons again
6. generates final answer

### 📌 Core Idea

Traditional LLM:

```
Plain text
Only thinks.
```

ReAct model:

```
Plain text
Thinks + interacts with environment.
```

## 🔷 2. Why ReAct Became Important

Earlier LLMs:

- only generated text

But modern AI systems require:

- web search
- API calls
- calculators
- databases
- code execution

LLMs alone cannot reliably solve these tasks without tools.

So ReAct was introduced to combine:

🧠 **reasoning + actions**

## 🔷 3. ReAct Workflow

### 🧠 Standard ReAct Flow

Plain text

```
Thought → Action → Observation → Thought → Final Answer
```

### 🔍 Step-by-Step Meaning

| Step | Meaning |
|---|---|
| Thought | Analyze problem |
| Action | Use tool/API |
| Observation | Read result |
| Final Answer | Generate response |

## 🔷 4. Simple ReAct Example

### ❓ User Question

Plain text

```
What is Tesla stock price today?
```

### 🧠 Model Reasoning

Plain text

```
Thought:
I need latest stock data.
```

### ⚡ Action

Plain text

```
Action:
Use stock market API/web search.
```

### 👀 Observation

Plain text

```
Observation:
Tesla stock price = $XXX
```

## 🔷 5. Difference Between CoT and ReAct

| Chain-of-Thought (CoT) | ReAct |
|---|---|
| Only reasoning | Reasoning + actions |
| Internal thinking | External tool interaction |
| Static answers | Dynamic workflows |
| No environment interaction | Uses APIs/tools |

## 🔥 Important Understanding

### 🧠 CoT

= Think step-by-step

### 🧠 ReAct

= Think + Take Action

## 🔷 6. Why ReAct is Powerful

ReAct allows models to:

- retrieve latest data
- perform calculations
- search documents
- use APIs
- execute code
- interact with external systems

## 🔷 7. Tools Used in ReAct Systems

Modern AI systems can use tools like:

| Tool | Purpose |
|---|---|
| Calculator | Math operations |
| Web Search | Latest information |
| Python | Code execution |
| Database | Retrieve data |
| APIs | External systems |
| Browser Tools | Website interaction |

## 🔷 8. Real-Life ReAct Example

### 🏢 Travel Assistant

User asks:

Plain text

```
Find cheapest flight from Delhi to Dubai tomorrow.
```

### 🧠 AI Reasoning

Plain text

```
Thought:
Need live flight information.
```

## ⚡ Action

Plain text

```
Use flight search API.
```

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## 👀 Observation

Plain text

```
Received flight data.
```

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## 🧠 Reason Again

Plain text

```
Compare prices and timings.
```

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## ✅ Final Answer

Plain text

```
Cheapest flight is XYZ Airlines at ₹18,000.
```

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## 🔷 9. ReAct in Modern AI Systems (2026)

Modern reasoning models like:

- GPT-5 systems
- Claude reasoning models
- Gemini agentic systems

use:

- reasoning
- planning
- tool orchestration
- multi-step workflows

ReAct principles are foundational to these systems.

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## 🔷 10. What is Structured Prompting?

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### ✅ Definition

Structured Prompting means:

> Designing prompts in an organized and controll
> predictable outputs.

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### 🔍 Core Idea

Instead of vague instructions:

- use structured templates
- define output format
- specify constraints

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## 🔷 11. Why Structured Prompting is Important

In production AI systems:
random outputs are dangerous.

Applications need:

- consistency
- automation
- reliability
- machine-readable outputs

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## 🔷 12. Unstructured vs Structured Prompting

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### ❌ Unstructured Prompt

Plain text

```
Tell me about Apple company.
```

Output may vary every time.

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### ✅ Structured Prompt

Plain text

```
Summarize Apple company in JSON format with:
- founder
- CEO
- headquarters
- products
```

Now output becomes:

- predictable
- structured
- automation-friendly

## 🔷 13. Components of Structured Prompting

| Component | Purpose |
| --- | --- |
| Role | Define AI identity |
| Instruction | Define task |
| Context | Background info |
| Constraints | Rules/limits |
| Output Format | Expected structure |

## 🔷 14. Role Prompting

### ✅ Definition

Assigning a role/persona to the model.

### 🔍 Example

Plain text

```
You are a senior financial analyst.
```

### 🧠 Why Useful?

Helps model:

- adapt tone
- use domain knowledge
- improve context understanding

## 🔷 15. Output Formatting

Modern AI systems often require outputs in:

- bullet points
- tables
- markdown
- JSON
- XML

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### 🔍 Example

Plain text

```
Return answer in JSON format only.
```

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## 🔷 16. Prompt Constraints

Constraints improve reliability.

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### 🔍 Examples

Plain text

```
Answer in 3 bullet points only.
```

Plain text

```
Use beginner-friendly English.
```

Plain text

```
Do not exceed 100 words.
```

## 🔷 17. Why Structured Prompting Matters in APIs

Production systems require:

- parsable outputs
- reliable formatting
- consistent responses

Without structure:
automation breaks.

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## 🔷 18. JSON Prompting

VERY IMPORTANT modern topic.

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### 🔍 Example

Plain text

```
Extract user details in JSON format:
{
 "name": "",
 "email": "",
 "phone": ""
}
```

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### 🧠 Why Important?

Used in:

- APIs
- workflows
- automation systems
- enterprise AI pipelines

## 🔷 19. Common Problems Without Structured Prompting

Without structure:

- inconsistent output
- hallucinated fields
- broken JSON
- automation failure

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## 🔷 20. Structured Prompting Best Practices

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### ✅ Be Specific

Bad:

Plain text

```
Explain AI.
```

Good:

Plain text

```
Explain transformers for beginners in 5 bullet points.
```

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### ✅ Define Output Format

Specify:

- JSON
- markdown
- table
- bullets

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## 🔷 21. Real-World Applications

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### 🏢 Customer Support Systems

Structured ticket generation.

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### 🏢 Financial AI

Structured report analysis.

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### 🏢 Medical AI

Standardized patient summaries.

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### 🏢 AI Automation Systems

Machine-readable outputs.

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## 🔷 22. ReAct + Structured Prompting Together

Modern AI systems combine:

- reasoning
- tool usage
- structured outputs

Example:

1. reason
2. use tool
3. retrieve data
4. return structured JSON

This is core architecture of modern AI products.

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